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Faulkner

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(54) **PROMOTION OF MEETING ENGAGEMENT
BY TRANSITIONING VIEWING
PERSPECTIVES TO A TEMPORARY
VIEWING PERSPECTIVE SHOWING GROUP
ACTIVITY**

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H04L 65/403 (2022.01)
H04N 7/15 (2006.01)

(52) **U.S. Cl.**
CPC **G06T 15/20** (2013.01); **H04L 65/403**
(2013.01); **H04N 7/157** (2013.01)

(58) **Field of Classification Search**
CPC G06T 15/20; H04L 65/403; H04N 7/157
See application file for complete search history.

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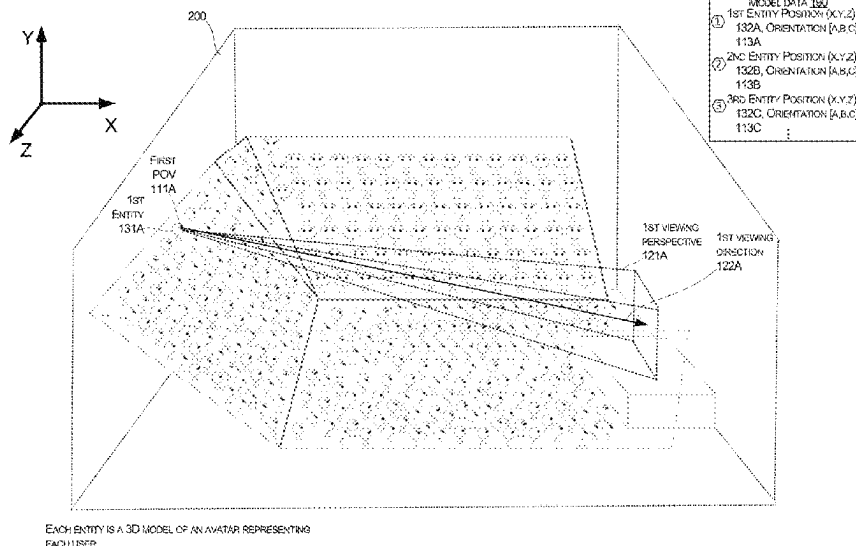
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(57) **ABSTRACT**

The techniques disclosed herein provide promotion of meet-
ing engagement by transitioning viewing perspectives to a
temporary viewing perspective showing group activity. A
system can show each person a view of a large virtual
environment, e.g., in a stadium full of representations of
meeting attendees. Each person sees the virtual environment
from a point of view originating from each person's repre-
sentation, e.g., a first-person avatar view. When a group
activity meets one or more conditions, the system generates
a new virtual environment model that shows detailed view
of all people in a group, without showing members of other
teams that may be intermingled with the group in an original
environment. The system may transition each group mem-
ber's view from the first-person view to a temporary view of
the newly generated model that only includes group mem-
bers. The temporary view can remain until the group activity
drops below a threshold.

20 Claims, 9 Drawing Sheets



MODEL DATA 190	
①	1ST ENTITY POSITION (X,Y,Z) 132A, ORIENTATION [A,B,C] 113A
②	2ND ENTITY POSITION (X,Y,Z) 132B, ORIENTATION [A,B,C] 113B
③	3RD ENTITY POSITION (X,Y,Z) 132C, ORIENTATION [A,B,C] 113C
⋮	

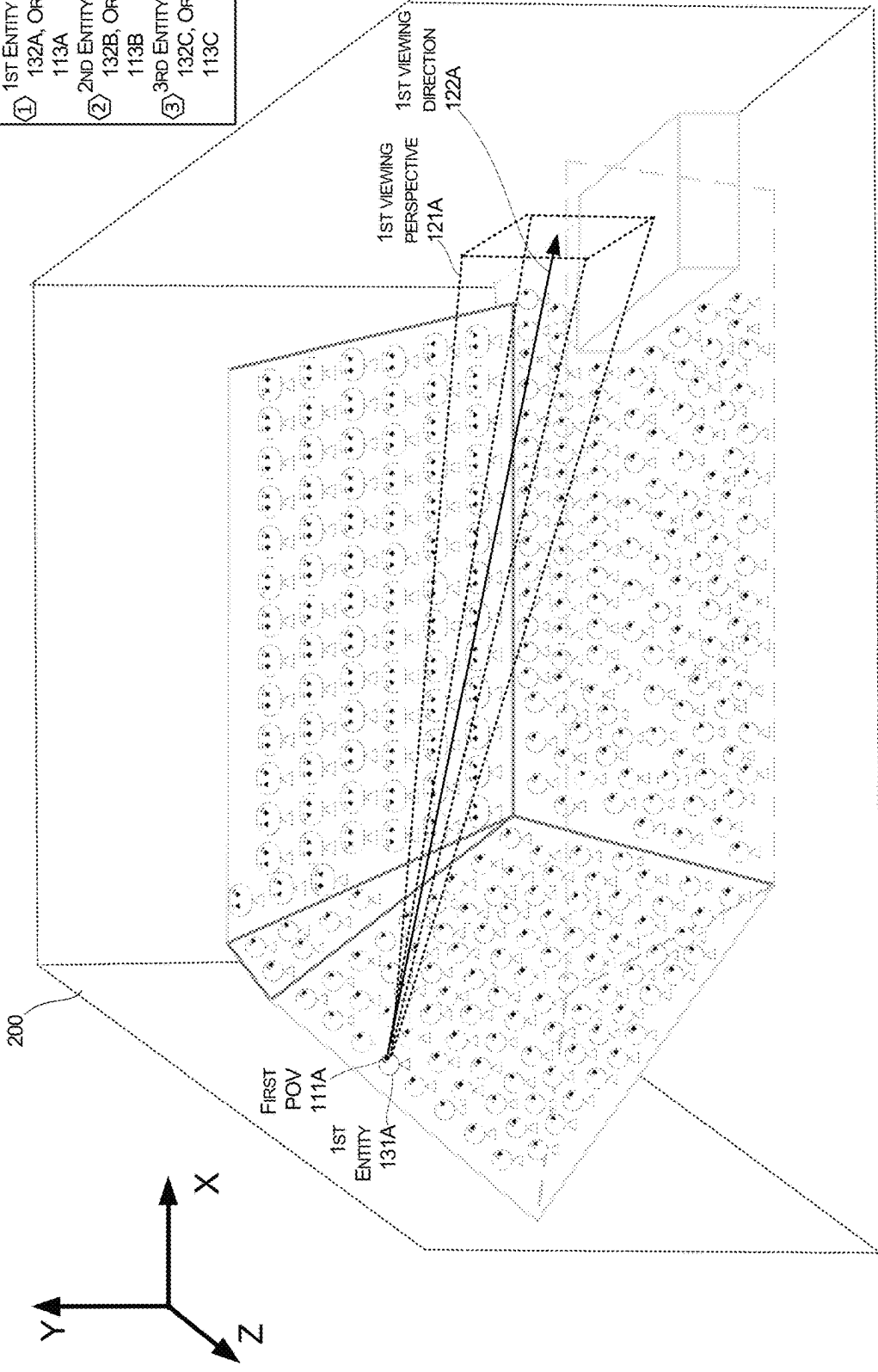


FIGURE 1

EACH ENTITY IS A 3D MODEL OF AN AVATAR REPRESENTING EACH USER

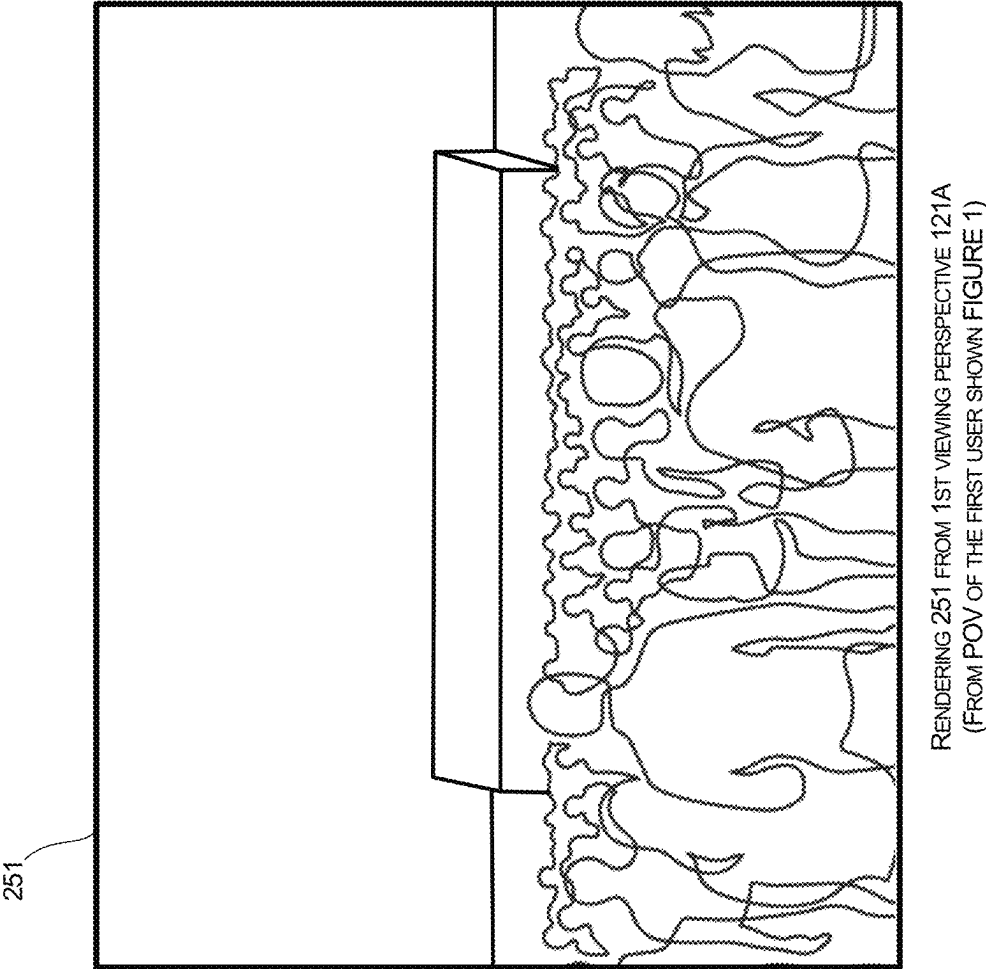


FIGURE 2

A THRESHOLD NUMBER OF USERS OF A
PREDETERMINED GROUP PERFORMS GESTURES

MODEL DATA 190	
①	1ST ENTITY POSITION (X,Y,Z) 132A, ORIENTATION [A,B,C] 113A
②	2ND ENTITY POSITION (X,Y,Z) 132B, ORIENTATION [A,B,C] 113B
③	3RD ENTITY POSITION (X,Y,Z) 132C, ORIENTATION [A,B,C] 113C
⋮	

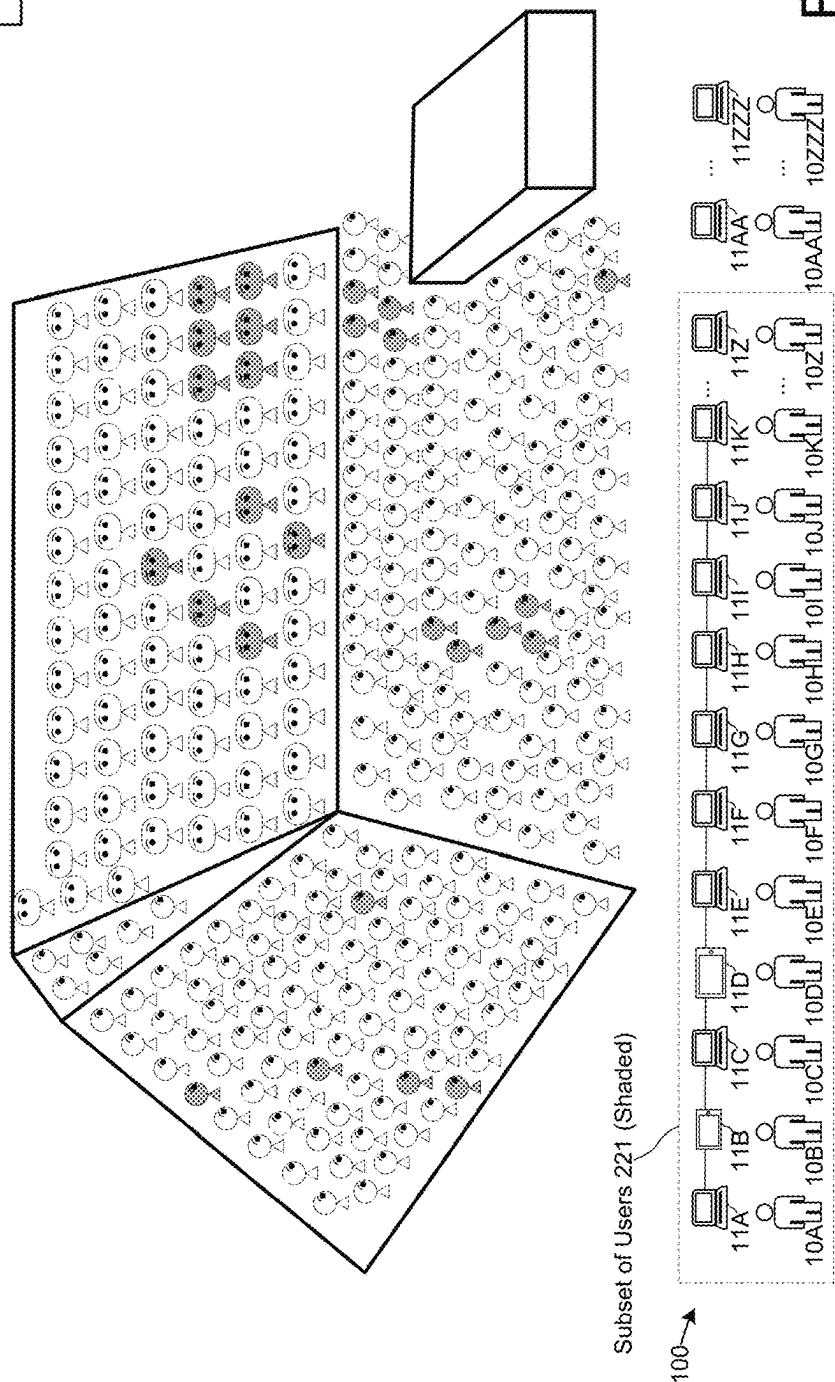


FIGURE 3

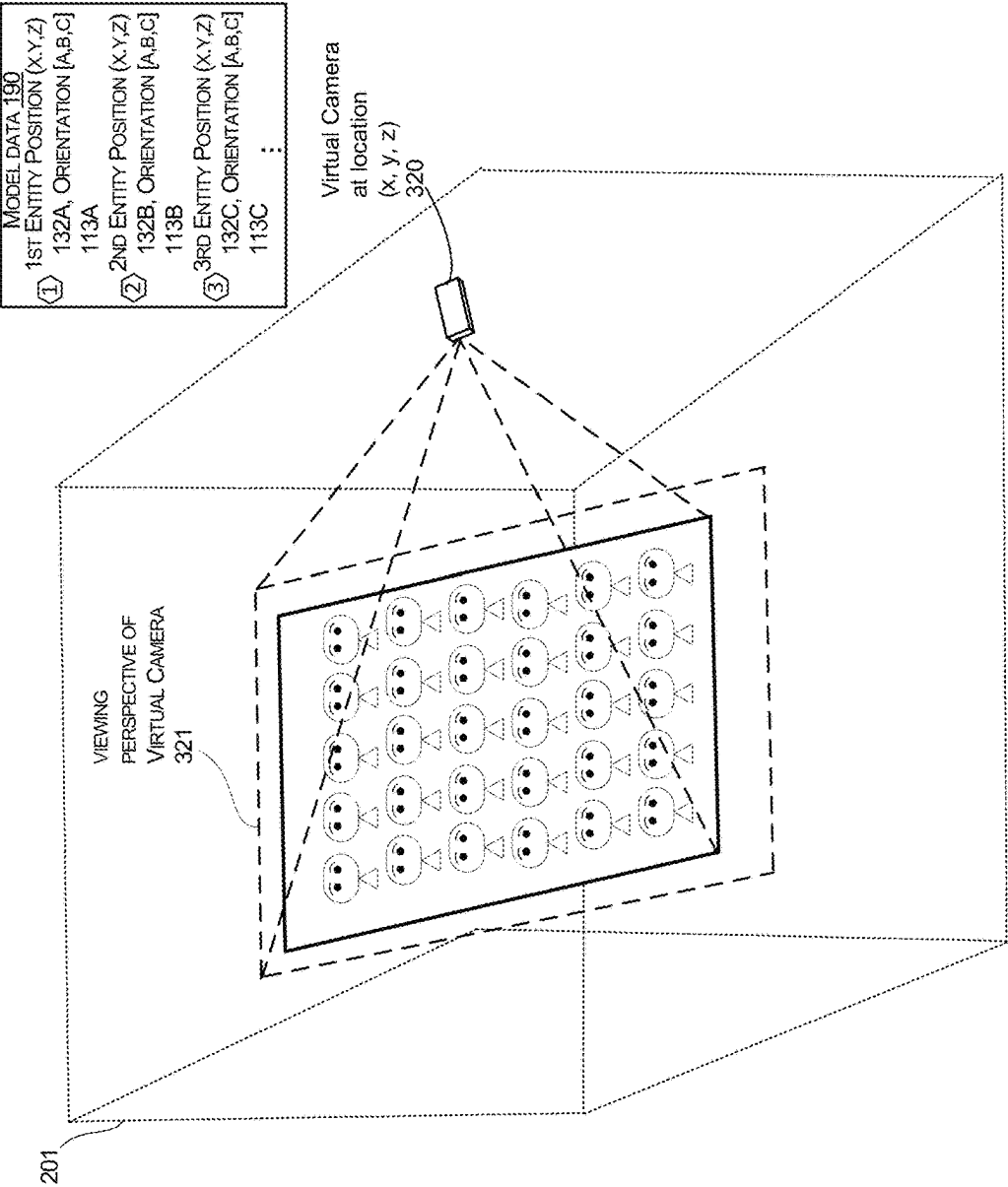
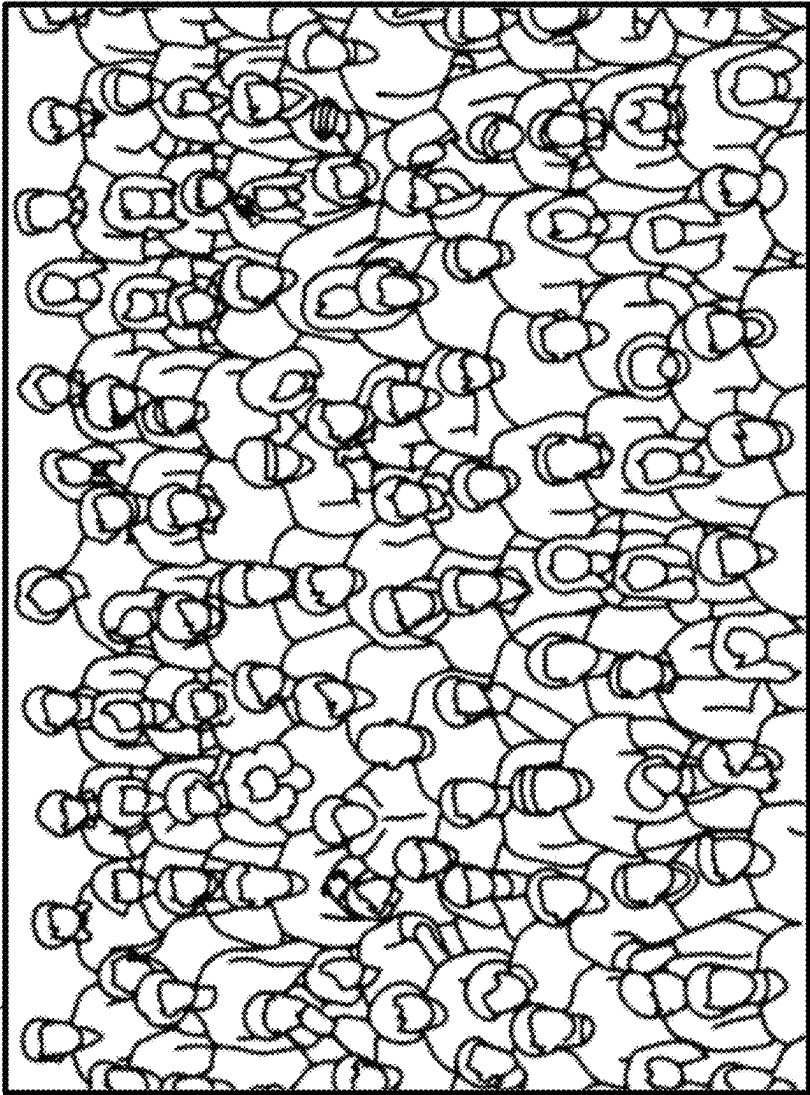


FIGURE 4

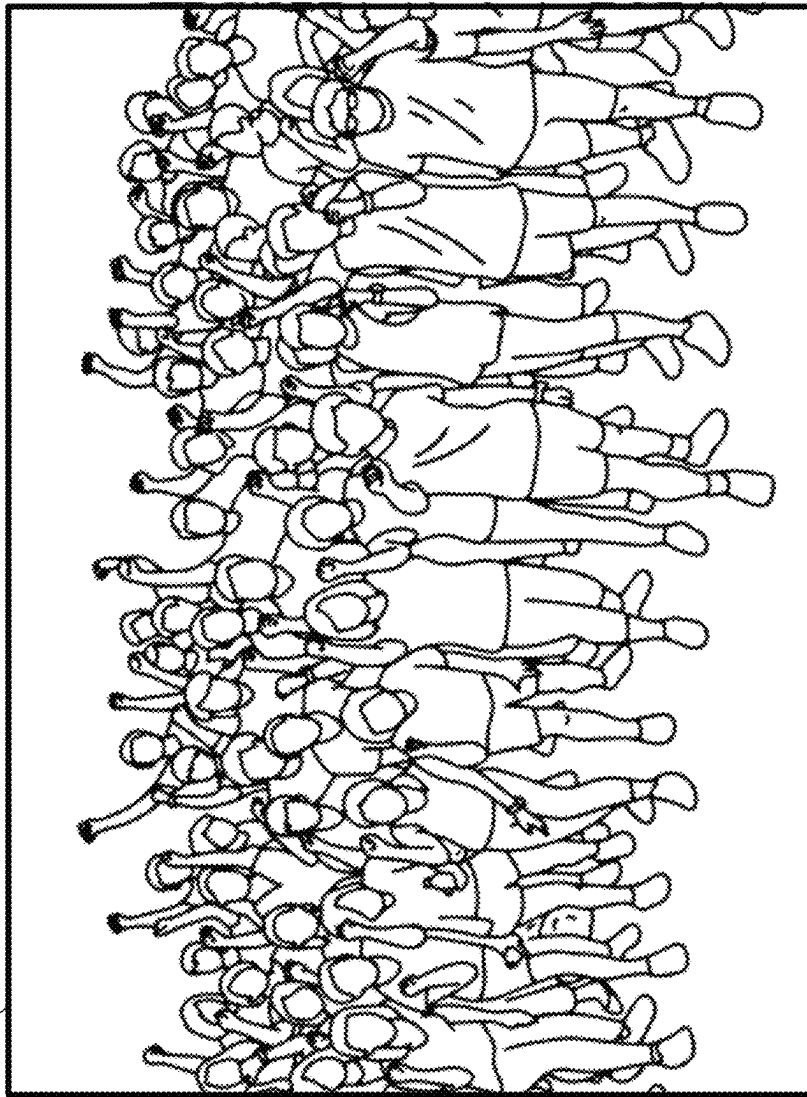
252



RENDERING 252 FROM VIEWING PERSPECTIVE OF VIRTUAL CAMERA 321
FROM SUPP MODEL 191
(DISPLAYED ON THE COMPUTER OF THE FIRST USER)

FIGURE 5

252



RENDERING 252 FROM VIEWING PERSPECTIVE OF VIRTUAL CAMERA 321
FROM SUPP MODEL 191
(DISPLAYED ON THE COMPUTER OF THE FIRST USER)

FIGURE 6

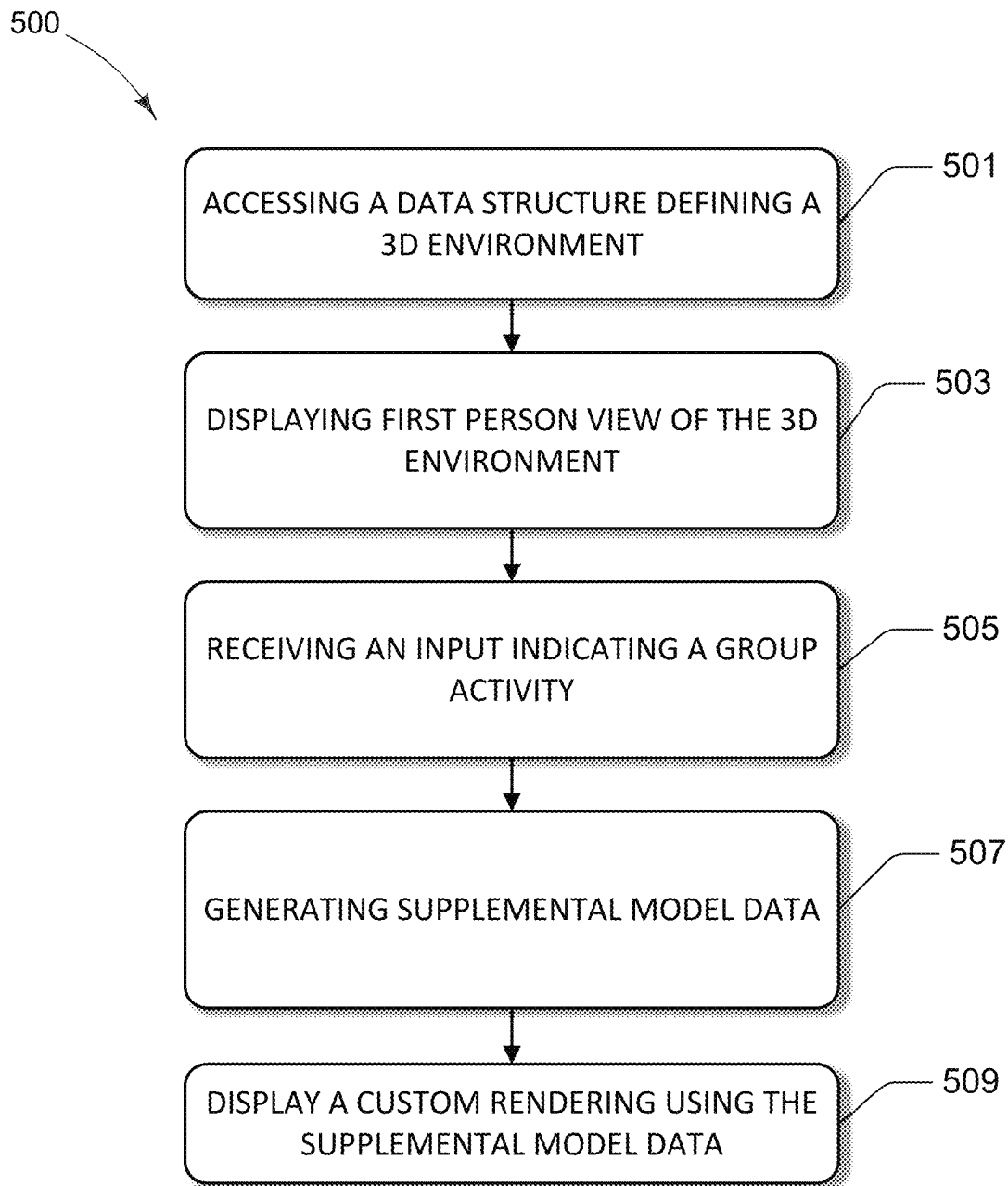
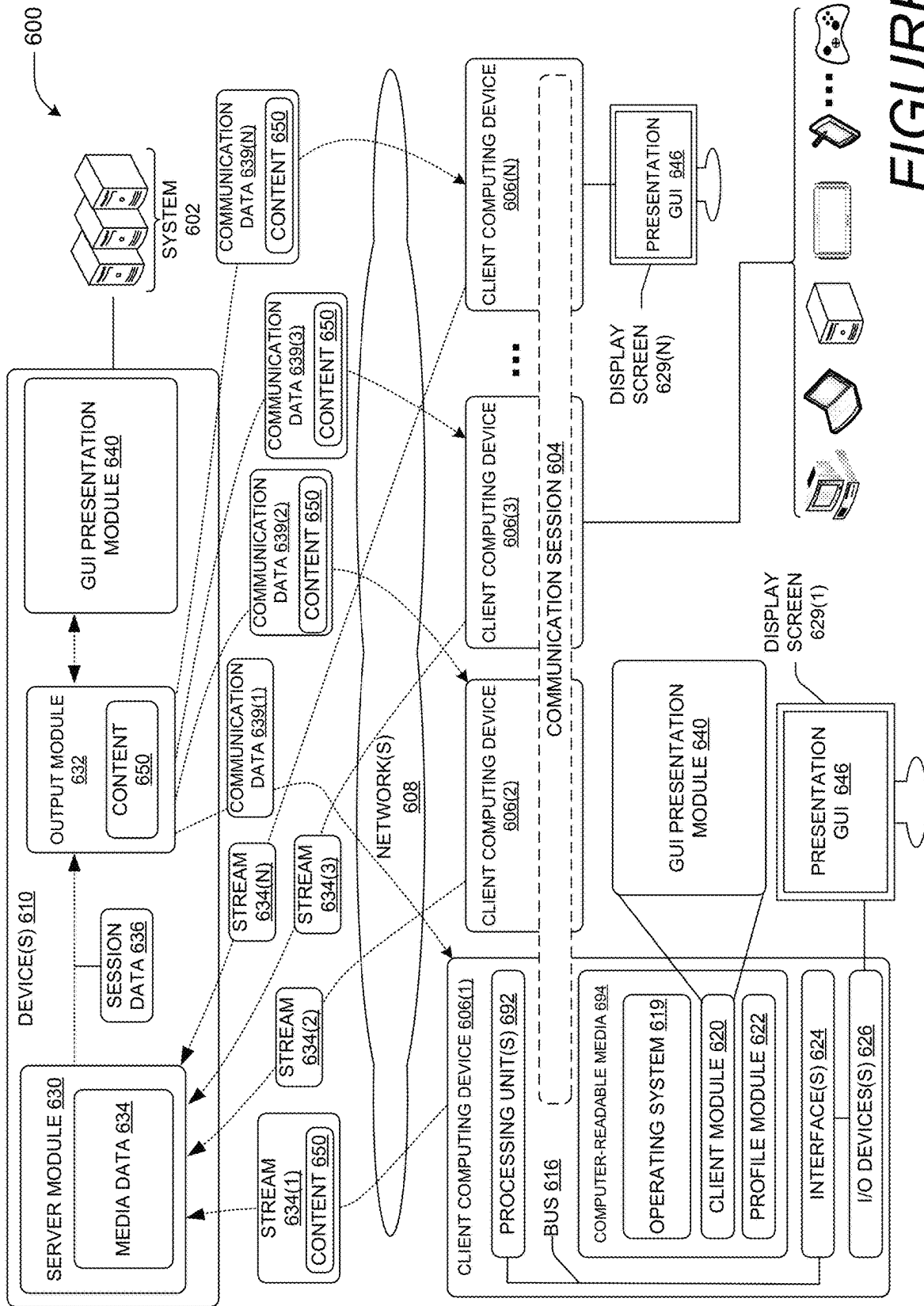


FIGURE 7



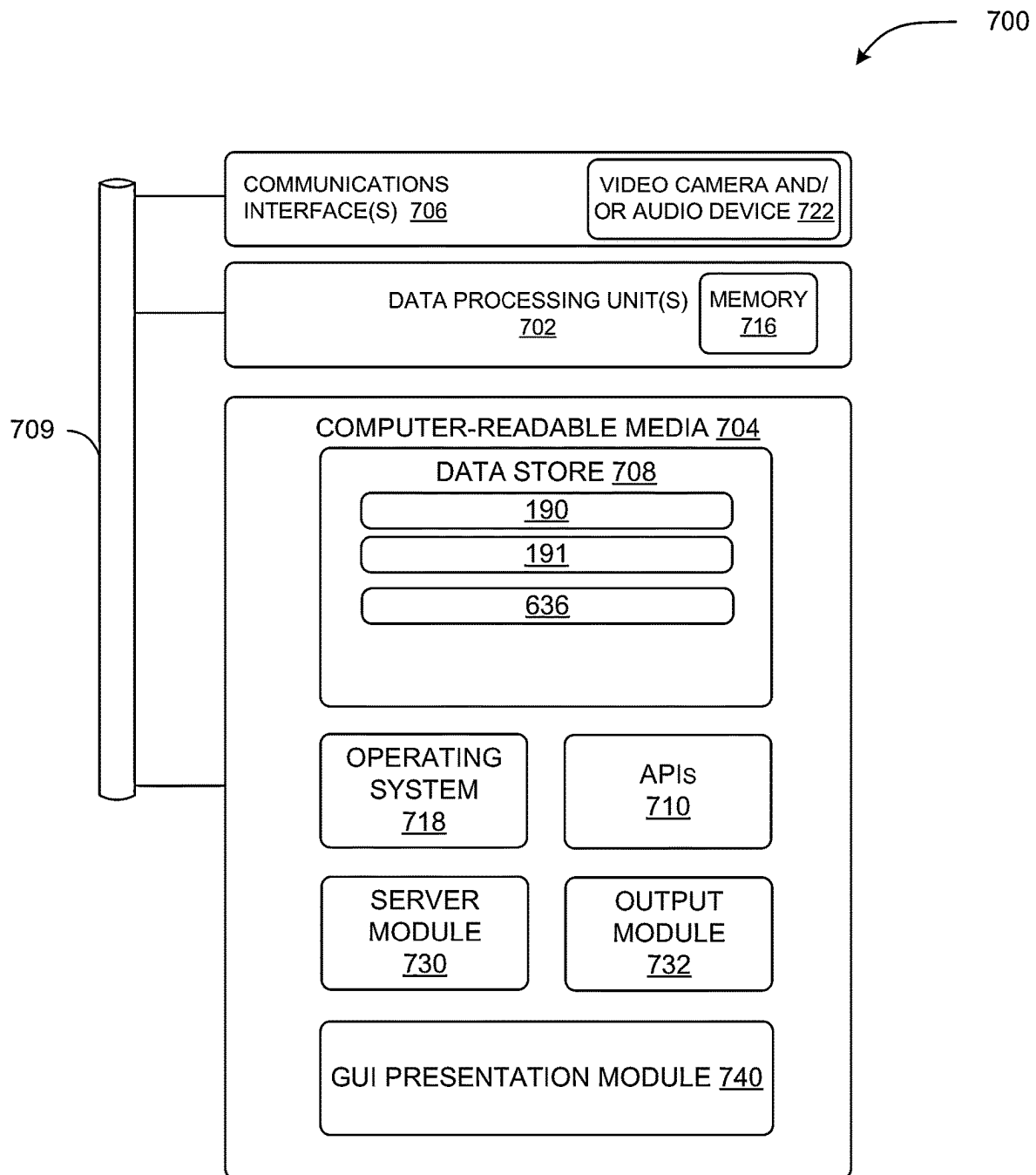


FIGURE 9

**PROMOTION OF MEETING ENGAGEMENT
BY TRANSITIONING VIEWING
PERSPECTIVES TO A TEMPORARY
VIEWING PERSPECTIVE SHOWING GROUP
ACTIVITY**

BACKGROUND

The use of meta-verse environments for on-line meeting applications is becoming ubiquitous. Participants of online meetings now meet in three-dimensional virtual environments and share content within those virtual environments. Despite a number of benefits over other forms of collaboration, the use of 3D environments for sharing content can raise a number of drawbacks.

When meeting in 3D immersive virtual environments, depending on positions of users and a virtual camera position, it may be hard to see the activity of specific groups of people. For instance, if a virtual environment has a large number of meeting attendees, not all members of particular group of people can be positioned near one another. This may occur when people are on several teams and each team may react differently to different events. For example, if a person is on a developer team, and that person has an avatar positioned with avatars of other team members, they can react together to certain events, e.g., when their team gets an award. However, if that person is also involved in a special project, e.g., on a volunteer group that involves a different group of people, that person cannot get the same group experience if that volunteer group experiences a similar event, e.g., receives an award. Since that person has an avatar that is positioned with the developer team, that person cannot react, e.g., cheer, and experience the same synergistic group experience he or she is experiencing with the developer team.

By not being close to team members in a virtual environment, users may not share the same experience, or benefit from a team-building experience. The users may miss salient information and gestures, which can lead to a loss of communication of information and enthusiasm. These shortcomings can lead to ineffective interactions between a computing device and a user. In addition, the above-described shortcomings of existing systems can lead to a loss in user engagement. In addition, proper control of permissions managing the views of a virtual environment is also important for securing proprietary or confidential information. This is especially important for events where competing teams are participants in a virtual meeting and different views are generated for each person.

Computing devices that do not promote user engagement, or worse, contribute to a loss of user engagement, can lead to production loss and inefficiencies with respect to a number computing resources. For instance, when a user becomes fatigued or disengaged, that user may need to refer to other resources, such as documents or participate in other forms of communication, when shared content is missed or overlooked. Missed content may need to be re-sent when viewers miss salient points or cues during a live meeting. Such activities can lead to inefficient or duplicative use of a network, processor, memory, or other computing resources. Thus, there is an ongoing need to develop improvements to help make the user experience more like an in-person meeting and more engaging. In addition, when information in meta-verse environments is not displayed correctly, some content that is supposed to be hidden from some users can be inadvertently exposed when such content is displayed on virtual objects, such as a virtual screen.

SUMMARY

The techniques disclosed herein provide promotion of meeting engagement by transitioning viewing perspectives to a temporary viewing perspective showing group activity. A system can show each person a view of a large virtual environment, e.g., in a stadium full of representations of meeting attendees. Each person sees the virtual environment from a point of view originating from each person's representation, e.g., a first-person avatar view. When a group activity meets one or more conditions, the system generates a new virtual environment model that shows detailed view of all people in a group, without showing members of other teams that may be intermingled with the group in an original environment. The system temporarily assembles a group to show the group activity. An input identifying an event that meets a preset condition can include people having a "threshold level of affiliation," e.g., an org chart shows they are team members, and/or people having a "threshold level of activity," e.g., 80% of the people on the team are cheering or providing a reaction or action input. The system may transition each group member's view from the first-person view to a temporary view of the newly generated model that only includes group members. The model can be a live activity model to show how each person is reacting and moving. The temporary view of the group activity can remain until the group activity drops below a threshold level.

Different groups can be assembled for a person's view throughout a meeting. For example, if a person is on a company team that gets an award, all the people of that team are assembled for that person's view to see the team's reaction. Then later, if that person gets an award for a special project that involves another group of people, that second group can be temporarily assembled for that person's view to see the group's reaction. Each group can be based on metadata such as an organizational chart or other data, such as a word processing document that multiple people made contributions to. Groups are selected based on a common activity or association. Certain group activity triggers system to assemble groups of people, like a company team, or group of workers for a special project. A threshold level of association for each person, e.g., did they work on a threshold number of documents together, or does an org chart show a relationship. For instance, a group may be people that are one level away from each other in a management org chart, where people outside of that parameter are not in that group.

The system provides fully inclusive group assembly that shows live activity of each person's avatar. This has more technical benefits than stitching images, as this disclosure puts the avatars together to make them look like they are positioned together in a group. This shows all individuals of a group the synergy of a current reaction to an event. Avatars can interact, e.g., high-five each other, look at one another, etc. The view can be from each person's first-person view in the new model or the view can be from a virtual camera.

In one example, a virtual stadium can have two different groups that are randomly intermingled. For instance, Seahawks and Falcons fans can have avatars that are intermingled. When Seahawks fans generate a level of synergy, e.g., they see an event in a game or react to the event, the system can take the avatars of the Seahawks fans that are scattered throughout different groups and put them in a single group to emphasize the group activity. The same can be done for fans of the other team. This means that each group does not have members of other teams inter-mixed in their group, e.g., a result that can lead to less than optimal

team reaction. The view of each person can be from the perspective originating from their avatar or from a virtual camera view showing the entire group.

This system allows for simultaneous views of different groups of people to show each group of people the synergistic activity of a team, project members, etc. For instance, a meeting may have thousands of avatars in an original virtual environment. A presenter may give an award to a group of people that participated in Project X. The techniques of the present disclosure can place avatars of the people that participated in Project X in a temporary model of a separate virtual showing only this first group of people, so they can react as a group to the announcement. The presenter can then give an award to a group of people that participated in Project Y. The techniques of the present disclosure can place avatars of the people that participated in Project Y in another temporary model of a separate virtual showing only this first group of people, so they can react as a group to the announcement. Each group can see a perspective view of the individual groups or they can see each other through a point of view originating from each avatar.

The adaptive adjustments using temporary views can provide a number of technical benefits for computing systems. For instance, by providing adaptive adjustments of perspective views, each user of a communication session can gain the benefit of group activity even though they are not sitting together in an original virtual environment. By providing this stimulus, a system can promote user engagement help a system to reduce user fatigue. By reducing user fatigue, particularly in a communication system, users can more effectively exchange information. This helps mitigate occurrences where shared content is missed or overlooked. This can reduce occurrences where users need to re-send information. More effective communication of shared content can also help avoid the need for external systems, such as mobile phones for texting and other messaging platforms. The system and features described herein can also help reduce the duplicative use of network, processor, memory, or other computing resources.

The features disclosed herein also provide a number of improvements for the security of a system. For example, members of other teams are rejected from seeing synergistic activity of other groups. This allows each group to share information and to communicate with one another without sharing that information with members of other groups. For instance, in the sporting event example described above, the first group of fans may react together and share information such as future strategies without allowing the other fans to see that information.

Features and technical benefits other than those explicitly described above will be apparent from a reading of the following Detailed Description and a review of the associated drawings. This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The term “techniques,” for instance, may refer to system(s), method(s), computer-readable instructions, module(s), algorithms, hardware logic, and/or operation(s) as permitted by the context described above and throughout the document.

BRIEF DESCRIPTION OF THE DRAWINGS

The Detailed Description is described with reference to the accompanying figures. In the figures, the left-most

digit(s) of a reference number identifies the figure in which the reference number first appears. The same reference numbers in different figures indicate similar or identical items. References made to individual items of a plurality of items can use a reference number with a letter of a sequence of letters to refer to each individual item. Generic references to the items may use the specific reference number without the sequence of letters.

FIG. 1 shows a virtual environment that includes representations of attendees of a meeting.

FIG. 2 shows an example of a rendering of a perspective view from a point of view that originates from a representation of one attendee.

FIG. 3 shows a highlight of meeting attendees that are associated with a particular event.

FIG. 4 shows an example of a supplemental model that is generated from an original model of a virtual environment, where the supplemental model includes a subset of meeting attendees that are associated with a particular event.

FIG. 5 shows an example of a rendering of a perspective view from a point of view that originates from a virtual camera that is defined in the supplemental model, where the attendees shown in the rendering are position within a virtual environment that shows aspects of an original virtual environment.

FIG. 6 shows an example of a rendering of a perspective view from a point of view that originates from a virtual camera that is defined in the supplemental model, where the attendees shown in the rendering are position within a new virtual environment.

FIG. 7 shows a flow diagram showing aspects of a routine that enables the disclosed techniques.

FIG. 8 is a computer architecture diagram illustrating an illustrative computer hardware and software architecture for a computing system capable of implementing aspects of the techniques and technologies presented herein.

FIG. 9 is a computer architecture diagram illustrating a computing device architecture for a computing device capable of implementing aspects of the techniques and technologies presented herein.

DETAILED DESCRIPTION

FIGS. 1-6 illustrate aspects of a process for promoting engagement by transitioning to a temporary viewing perspective from a first viewing perspective associated with a first entity that is one of a plurality of entities positioned within a virtual environment. FIG. 1 shows aspects of virtual environment 200 that includes a number of entities 131A-131ZZ. In this example, the entities are in the form of avatars, which are also referred to herein as representations of users. For example, the first entity first entity 131A is in the form of an avatar that represents a first participant of a communication session. As shown, each entity has a location in the 3D environment 200, which may be based on coordinates, and each entity has an orientation, which is represented as a vector originating from a point on the entity, such as a point of view that is at the location of an avatar's eyes. The point of view may also be associated with a location in the 3D environment. In some configurations, an orientation vector 122 can indicate which direction an entity is facing. In this example, the first orientation vector 122A shows the direction that the first entity 131A is facing, and the first orientation vector 122A is based on a first point of view 111A of the first entity. The orientation vector for an entity can also be referred to as an “original orientation” of the entity that is based on the original model data 190.

Also shown, model data **190**, also referred to herein as a first model **190**, the original model data **190** or a data structure **190**, can define positions and orientations for a plurality of entities **131**. An entity can be a virtual object such as an avatar representing a user or an entity can be another type of virtual object, such as a virtual television screen, virtual furniture, a virtual camera, etc. A virtual camera is similar to an avatar in that it may have a position, orientation and a viewing perspective but it may not have a displayable entity, e.g., it may be an invisible virtual camera.

The model data **190** can also define viewing perspectives **121** that originate from a point of view **111** for an individual entity. The viewing perspectives **121** extend in a direction with the orientation vector and each viewing perspective **121** can have dimensions, such as a width or height, that increase in size as a distance from the point of view increases. The model data can also define physical characteristics of each entity, such as size, shape, surface patterns, etc.

In one example, an entity can be an avatar or representation of a user. The point of view **111** can be at a specific location relative to an entity, such as a point of where the eyes of an avatar are located. The viewing perspective **121** of an entity can project from the point of view **111**, and the viewing perspective **121** can be used to generate a rendering that is displayed to the user controlling the avatar. The system can display entities of a 3D environment **200** based on the location and orientation the avatar as well as the position of entities that are positioned within the viewing perspective **121**. In the example of FIG. 1, the first entity **131A** has a first viewing perspective **121A** that follows the direction of the orientation vector **122A**.

As shown in FIG. 2, a rendering **251** of the first viewing perspective **121A** can include any entity that is within the viewing perspective **121A**, each avatar within the boundaries of the first viewing perspective **121A**, and a portion of the stage of the virtual environment. A rendering **251** of a viewing perspective **121A** can be a 2D image that is generated using any suitable projection technique that captures images from the first point of view **111A**. A rendering generated from a viewing perspective may crop a portion of an entity that is not within the boundaries of the viewing perspective, as shown in FIG. 2. In this example, the system displays the first viewing perspective **121A** to the first user by generating a rendering of the 3D environment **200** using the first viewing perspective **121A** associated with the first entity **131A** for display on a display device **11A** associated with the first user **10A**. This view is generated by use of the model data **190**.

As shown in FIGS. 3-5, one or more detected events causes the system to transition to a rendering **252** of a temporary viewing perspective **321** from a first viewing perspective **121A** associated with a first entity **131A** that is one of a plurality of entities **131A-131ZZ** positioned within a 3D environment **200**. FIG. 3 shows aspects of a system **100** having computers **11** each associated with people **10** who are participants of a communication session, such as a meeting in a virtual environment **200**, which allows each person to communicate audio signals and allows each person to view each other through their own viewing perspectives originating from their avatars. FIG. 3 shows an example of a group of entities that are related to a particular event. These entities that are related to an event meeting a preset condition are shaded to illustrate aspects of this example.

While the system causes the display of the rendering **251** of a first viewing perspective **121A** to the first computer **11A**, the system can also do the same for each person having a representation in the virtual environment **200**. While in this

viewing state, the system may receive signals from users indicating an event. The signals may be a computer-based input, such as an input causing the display of a reaction, e.g., the posting of a reaction emoji, or the signals may be an audio signal or a text signal indicating a specific event. For example, a presenter may indicate in an announcement that Team X won an award. When the system detects such an event that is related to a particular set of users, e.g., Team X, the system can retrieve organizational data or other metadata to determine identifiers of each user that's part of Team X. As shown in FIGS. 3 and 4, in response to determining that the input identifying the event meets the preset condition, the system can then use those identifiers to generate a supplemental model **191** (also referred to herein as a "second model **191**") defining a second 3D environment **201** comprising individual entities **131A-131Z** (also referred to herein as the subset of individual entities **131A-131Z**) each representing the users **10A-10Z** associated with the event, wherein the individual entities **131A-131Z** are positioned within a temporary viewing perspective **321** of a virtual camera **320**. In this example, a subset **221** of users **10A-10Z** are related to the event from an input they provided or by having a threshold level of affiliation. The supplemental model can have positions and orientations of each entity.

The size and shape of the temporary viewing perspective can be based on the number of entities in a group. For example, parameters of the temporary viewing perspective **321** can include coordinates in the virtual environment. The coordinate can be configured such that the border of the temporary viewing perspective **321** is around a cluster of selected entities meeting the criteria defined herein. The distance between the border of the temporary viewing perspective **321** and the virtual camera **320** can be based on the density of the entities, the number of entities, and the shape of the cluster of select entities. The position of the virtual camera can include coordinates that position the virtual camera such that a particular surface of the entities, e.g., the face of the avatars, is displayed in a rendering generated from the temporary viewing perspective **321**.

As shown in FIG. 5, in response to determining that the input identifying the event meets the preset condition, the system can cause the display of a rendering that is based on the supplemental model **191**. In this state of the system, the system can generate a rendering **251** of the individual entities **131A-131Z** positioned within the temporary viewing perspective **321** of the virtual camera **320** for display on a screen of a computer **11A** associated with the first user **10A**. The rendering of the individual entities are displayed using the supplemental model **191** defining the second 3D environment **201** comprising individual entities **131A-131Z** each representing the users **10A-10Z** associated with the event.

Users can be associated with the event and the event can meet a preset condition using one or more factors. For example, a system can receive signals from individual computers **11**, each person can be making a sound, e.g., cheering, or providing a signal, such as a cheering emoji. users of an event. The system can determine that an event meets a preset condition when each user providing the signal has a threshold level of affiliation, e.g., an org chart shows that they are members of a corporate team, and when each users having the threshold level of affiliation have a threshold level of activity. A threshold level of activity can be when more than 80% of a team is cheering or providing an active input. Thus, when less than a threshold level of activity is present, e.g., using an example threshold of 80%, when less than 80% of a team is cheering, the system the

event does not meet the preset condition and the system does not generate parameters of the temporary viewing perspective 321.

This allows the system to generate models for each team that meets a threshold. This means that a person can be members of different teams or different projects and they can see groups of people that belong to each team or project reacting together. For instance, if a person is part of a developer team and a threshold number of those developers cheer, that person can see themselves in a cluster of avatars with team members. If that same person worked on a multi user editing session of a spreadsheet, and people were recognized in an announcement regarding that spreadsheet, the system can generate a temporary model that shows all of the avatars of those contributors in a group reacting together.

Thus, in some embodiments, the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices 11A-11Z associated with a subset 221 of users 10A-10Z have a threshold level of affiliation with one another, and where the activity signals indicate the threshold level of activity, wherein the generation of the supplemental model 191 and the rendering of the individual entities 131A-131Z positioned within the temporary viewing perspective 321 of the virtual camera 320 is in response to receiving the activity signals indicating the threshold level of activity from the subset of computing devices 11A-11Z associated with the subset 221 of users 10A-10Z having the threshold level of affiliation with one another

In some configurations, the event may meet a preset condition based on affiliations, without the use of activity signals. Thus, in some embodiments, the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices 11A-11Z associated with a subset 221 of users 10A-10Z have a threshold level of affiliation with one another, wherein the generation of the supplemental model 191 and the rendering of the individual entities 131A-131Z positioned within the temporary viewing perspective 321 of the virtual camera 320 is in response to receiving the activity signals from the subset of computing devices 11A-11Z associated with the subset 221 of users 10A-10Z having the threshold level of affiliation with one another.

In some configurations, the event may meet a preset condition based on activity levels, without the levels of affiliation. Thus, in some embodiments, the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices 11A-11Z associated with a subset 221 of users 10A-10Z and the activity signals indicate a threshold level of activity, e.g., they are providing a predetermined volume level or providing a predetermined input, e.g., a particular emoji posting. Thus, the generation of the supplemental model 191 and the rendering of the individual entities 131A-131Z positioned within the temporary viewing perspective 321 of the virtual camera 320 is in response to receiving the activity signals from the subset of computing devices 11A-11Z associated with the subset 221 of users 10A-10Z, wherein the activity signals of the subset 221 of users 10A-10Z indicate the threshold level of activity.

In some embodiments, the rendering of the individual entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset 221 of users reduces below a second activity threshold level. Thus, in response to the activity signals of the subset 221 of users reducing below the second activity threshold level, the display on the display

device 11A associated with the first user 10A returns to the rendering of the 3D environment 200 using the first viewing perspective 121A associated with the first entity 131A. This can apply to all users of a group.

The system can also modify permissions in response to the input identifying the event meets the preset condition. For example, the modified permission allow a first set of computing devices of the users 10A-10Z associated with the event to display the rendering of the individual entities using the supplemental model 191 defining the second 3D environment 201. The permissions are also modified in response to the event to restrict a second set of computers associated with other users from displaying the rendering of the individual entities using the supplemental model 191 defining the second 3D environment 201.

The example rendering of FIG. 5 shows an example of a temporary virtual environment 201 that utilize aspects of the original virtual environment 200. For instance, if the original virtual environment has a stadium with seats, some of those stadium seats may be used in the supplemental model for the temporary virtual environment 201. However, as shown in FIG. 6, temporary virtual environment 201 may not use aspects of the original virtual environment 200. In this example, the entities are placed in a new virtual environment, e.g., standing on a flat surface.

FIG. 7 is a diagram illustrating aspects of a routine 500 for promoting engagement by transitioning to a temporary viewing perspective 321 from a first viewing perspective 121A associated with a first entity 131A that is one of a plurality of entities 131A-131ZZ positioned within a 3D environment 200. It should be understood by those of ordinary skill in the art that the operations of the methods disclosed herein are not necessarily presented in any particular order and that performance of some or all of the operations in an alternative order is possible and is contemplated. The operations have been presented in the demonstrated order for ease of description and illustration. Operations may be added, omitted, performed together, and/or performed simultaneously, without departing from the scope of the appended claims.

It should also be understood that the illustrated methods can start or end at any time and need not be performed in their entirety. Some or all operations of the methods, and/or substantially equivalent operations, can be performed by execution of computer-readable instructions included on a computer-storage media, as defined herein. The term "computer-readable instructions," and variants thereof, as used in the description and claims, is used expansively herein to include routines, applications, application modules, program modules, programs, components, data structures, algorithms, and the like. Computer-readable instructions can be implemented on various system configurations, including single-processor or multiprocessor systems, minicomputers, mainframe computers, personal computers, hand-held computing devices, microprocessor-based, programmable consumer electronics, combinations thereof, and the like. Although the example routine described below is operating on a system, e.g., one or more computing devices, it can be appreciated that this routine can be performed on any computing system which may include any number of computers working in concert to perform the operations disclosed herein.

Thus, it should be appreciated that the logical operations described herein are implemented as a sequence of computer implemented acts or program modules running on a computing system such as those described herein and/or as interconnected machine logic circuits or circuit modules

within the computing system. The implementation is a matter of choice dependent on the performance and other requirements of the computing system. Accordingly, the logical operations may be implemented in software, in firmware, in special purpose digital logic, and any combination thereof.

Additionally, the operations illustrated in FIG. 7 and the other FIGURES can be implemented in association with the example user interfaces and systems described herein. For instance, the various devices and/or modules described herein can generate, transmit, receive, and/or display data associated with content of a communication session e.g., live content, broadcasted event, recorded content, etc. and/or a presentation UI that includes renderings of one or more participants of remote computing devices, avatars, channels, chat sessions, video streams, images, virtual objects, and/or applications associated with a communication session.

The routine 500 includes an operation 501 where the system 100 accesses a data structure 190 defining the 3D environment 200, the data structure 190 defines a position 132 and an orientation 113 for each entity of the plurality of entities 131 of the 3D environment 200, the data structure 190 defines at least one viewing perspective 121 that is based on one or more points of view 111 for individual entities 131 that are selectable for generating custom renderings 300 of the at least one viewing perspective 121 for a computing device 11A of a communication session 604.

At operation 503, the system generates a rendering of the 3D environment 200 using the first viewing perspective 121A associated with the first entity 131A for display on a display device 11A associated with the first user 10A. This view is generated by the use of the original model data 190. FIG. 2 shows an example where the rendering includes entities that are positioned within the first viewing perspective 121A to the first user 10A.

At operation 505, the system determines that an input identifying an event meets a preset condition. An event can meet a preset condition based on one or more factors. For example, a system can determine that an event meets a preset condition when users having a threshold level of affiliation e.g., members of a corporate team, have a threshold level of activity, e.g., 80% of team is cheering. An event can also meet a preset condition by a single action as well, e.g., a presenter mentions that Team X has won an award. This way, if the presenter mentions that Team X and Team Y has each won awards, each team can see the team reacting in individual groups without having to see members of other teams, e.g., Team X members do not see Team Y members, and vice versa.

At operation 507, the system generates supplemental model data 191 defining at least one of a new position and a new orientation for entities associated with an event. This generation can be in response to an event meeting a preset condition. An example of the supplemental model data is shown in FIG. 4. The supplemental model data 191 is generated to create the new view just for the users associated with an event. The generation of the supplemental model data 191 does not cause a modification of the data structure 190 defining the 3D environment 200. In some embodiments, during the generation or modification of the supplemental model data, the position 132 and the orientation 113 for each entity of the plurality of entities 131 of the 3D environment 200 defined in the data structure 190 are not modified. System permissions can also restrict the modification of the data structure 190 during the modification or generation of the supplemental model data to preserve the other perspective views of the original model 190.

At operation 509, the system 100 causes a display of the custom rendering 251 for the computing device 11A using the supplemental model data 191 defining positions and orientations of entities associated with users of the detected event. This custom rendering 252 is based on the viewing perspective that originates from the point of view associated with the virtual camera defined in the supplemental model data 191. This view can also be from a point of view of an entity, such as an avatar of a user, defined in the supplemental model data 191. Each model can also define permissions for viewing. For example, the supplemental model data 191 can define permissions that only allow members of a particular team, or people associated with a particular event, to view any rendering generated by the supplemental model data 191.

In some embodiments, the system restricts the transition from the rendering (251) of the using the first viewing perspective (121A) to the rendering (252) of the viewing perspective of the supplemental model (191) until the subset (221) of users (10A-10Z) has the threshold level of affiliation with one another. This saves the system resources to do transitions until people having actual affiliations, e.g., they are on the same team in an organization chart, have a common working file each person made contributions to, or are part of a common project, are part of an event, such as an announcement about the subset of users or the subset of users provide a threshold level of activity, e.g., 80% of the subset of users provides an input showing an expression, e.g., an emoji or a message.

FIG. 8 is a diagram illustrating an example environment 600 in which a system 602 can implement the techniques disclosed herein. It should be appreciated that the above-described subject matter may be implemented as a computer-controlled apparatus, a computer process, a computing system, or as an article of manufacture such as a computer-readable storage medium. The operations of the example methods are illustrated in individual blocks and summarized with reference to those blocks. The methods are illustrated as logical flows of blocks, each block of which can represent one or more operations that can be implemented in hardware, software, or a combination thereof. In the context of software, the operations represent computer-executable instructions stored on one or more computer-readable media that, when executed by one or more processors, enable the one or more processors to perform the recited operations.

Generally, computer-executable instructions include routines, programs, objects, modules, components, data structures, and the like that perform particular functions or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described operations can be executed in any order, combined in any order, subdivided into multiple sub-operations, and/or executed in parallel to implement the described processes. The described processes can be performed by resources associated with one or more device(s) such as one or more internal or external CPUs or GPUs, and/or one or more pieces of hardware logic such as field-programmable gate arrays ("FPGAs"), digital signal processors ("DSPs"), or other types of accelerators.

All of the methods and processes described above may be embodied in, and fully automated via, software code modules executed by one or more general purpose computers or processors. The code modules may be stored in any type of computer-readable storage medium or other computer storage device, such as those described below. Some or all of the

methods may alternatively be embodied in specialized computer hardware, such as that described below.

Any routine descriptions, elements or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code that include one or more executable instructions for implementing specific logical functions or elements in the routine. Alternate implementations are included within the scope of the examples described herein in which elements or functions may be deleted, or executed out of order from that shown or discussed, including substantially synchronously or in reverse order, depending on the functionality involved as would be understood by those skilled in the art.

In some implementations, a system 602 may function to collect, analyze, and share data that is displayed to users of a communication session 604. As illustrated, the communication session 603 may be implemented between a number of client computing devices 606(1) through 606(N) (where N is a number having a value of two or greater) that are associated with or are part of the system 602. The client computing devices 606(1) through 606(N) enable users, also referred to as individuals, to participate in the communication session 603.

In this example, the communication session 603 is hosted, over one or more network(s) 608, by the system 602. That is, the system 602 can provide a service that enables users of the client computing devices 606(1) through 606(N) to participate in the communication session 603 (e.g., via a live viewing and/or a recorded viewing). Consequently, a “participant” to the communication session 603 can comprise a user and/or a client computing device (e.g., multiple users may be in a room participating in a communication session via the use of a single client computing device), each of which can communicate with other participants. As an alternative, the communication session 603 can be hosted by one of the client computing devices 606(1) through 606(N) utilizing peer-to-peer technologies. The system 602 can also host chat conversations and other team collaboration functionality (e.g., as part of an application suite).

In some implementations, such chat conversations and other team collaboration functionality are considered external communication sessions distinct from the communication session 603. A computing system 602 that collects participant data in the communication session 603 may be able to link to such external communication sessions. Therefore, the system may receive information, such as date, time, session particulars, and the like, that enables connectivity to such external communication sessions. In one example, a chat conversation can be conducted in accordance with the communication session 603. Additionally, the system 602 may host the communication session 603, which includes at least a plurality of participants co-located at a meeting location, such as a meeting room or auditorium, or located in disparate locations.

In examples described herein, client computing devices 606(1) through 606(N) participating in the communication session 603 are configured to receive and render for display, on a user interface of a display screen, communication data. The communication data can comprise a collection of various instances, or streams, of live content and/or recorded content. The collection of various instances, or streams, of live content and/or recorded content may be provided by one or more cameras, such as video cameras. For example, an individual stream of live or recorded content can comprise media data associated with a video feed provided by a video camera (e.g., audio and visual data that capture the appear-

ance and speech of a user participating in the communication session). In some implementations, the video feeds may comprise such audio and visual data, one or more still images, and/or one or more avatars. The one or more still images may also comprise one or more avatars.

Another example of an individual stream of live or recorded content can comprise media data that includes an avatar of a user participating in the communication session along with audio data that captures the speech of the user. Yet another example of an individual stream of live or recorded content can comprise media data that includes a file displayed on a display screen along with audio data that captures the speech of a user. Accordingly, the various streams of live or recorded content within the communication data enable a remote meeting to be facilitated between a group of people and the sharing of content within the group of people. In some implementations, the various streams of live or recorded content within the communication data may originate from a plurality of co-located video cameras, positioned in a space, such as a room, to record or stream live a presentation that includes one or more individuals presenting and one or more individuals consuming presented content.

A participant or attendee can view content of the communication session 603 live as activity occurs, or alternatively, via a recording at a later time after the activity occurs. In the examples described herein, client computing devices 606(1) through 606(N) participating in the communication session 603 are configured to receive and render for display, on a user interface of a display screen, communication data. The communication data can comprise a collection of various instances, or streams, of live and/or recorded content. For example, an individual stream of content can comprise media data associated with a video feed (e.g., audio and visual data that capture the appearance and speech of a user participating in the communication session). Another example of an individual stream of content can comprise media data that includes an avatar of a user participating in the conference session along with audio data that captures the speech of the user. Yet another example of an individual stream of content can comprise media data that includes a content item displayed on a display screen and/or audio data that captures the speech of a user. Accordingly, the various streams of content within the communication data enable a meeting or a broadcast presentation to be facilitated amongst a group of people dispersed across remote locations.

A participant or attendee to a communication session is a person that is in range of a camera, or other image and/or audio capture device such that actions and/or sounds of the person which are produced while the person is viewing and/or listening to the content being shared via the communication session can be captured (e.g., recorded). For instance, a participant may be sitting in a crowd viewing the shared content live at a broadcast location where a stage presentation occurs. Or a participant may be sitting in an office conference room viewing the shared content of a communication session with other colleagues via a display screen. Even further, a participant may be sitting or standing in front of a personal device (e.g., tablet, smartphone, computer, etc.) viewing the shared content of a communication session alone in their office or at home.

The system 602 of FIG. 8 includes device(s) 610. The device(s) 610 and/or other components of the system 602 can include distributed computing resources that communicate with one another and/or with the client computing devices 606(1) through 606(N) via the one or more network(s) 608. In some examples, the system 602 may be

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an independent system that is tasked with managing aspects of one or more communication sessions such as communication session 603. As an example, the system 602 may be managed by entities such as SLACK, WEBEX, GOTOMEETING, GOOGLE HANGOUTS, etc.

Network(s) 608 may include, for example, public networks such as the Internet, private networks such as an institutional and/or personal intranet, or some combination of private and public networks. Network(s) 608 may also include any type of wired and/or wireless network, including but not limited to local area networks (“LANs”), wide area networks (“WANs”), satellite networks, cable networks, Wi-Fi networks, WiMax networks, mobile communications networks (e.g., 3G, 4G, and so forth) or any combination thereof. Network(s) 608 may utilize communications protocols, including packet-based and/or datagram-based protocols such as Internet protocol (“IP”), transmission control protocol (“TCP”), user datagram protocol (“UDP”), or other types of protocols. Moreover, network(s) 608 may also include a number of devices that facilitate network communications and/or form a hardware basis for the networks, such as switches, routers, gateways, access points, firewalls, base stations, repeaters, backbone devices, and the like.

In some examples, network(s) 608 may further include devices that enable connection to a wireless network, such as a wireless access point (“WAP”). Examples support connectivity through WAPs that send and receive data over various electromagnetic frequencies (e.g., radio frequencies), including WAPs that support Institute of Electrical and Electronics Engineers (“IEEE”) 802.11 standards (e.g., 802.11g, 802.11n, 802.11ac and so forth), and other standards.

In various examples, device(s) 610 may include one or more computing devices that operate in a cluster or other grouped configuration to share resources, balance load, increase performance, provide fail-over support or redundancy, or for other purposes. For instance, device(s) 610 may belong to a variety of classes of devices such as traditional server-type devices, desktop computer-type devices, and/or mobile-type devices. Thus, although illustrated as a single type of device or a server-type device, device(s) 610 may include a diverse variety of device types and are not limited to a particular type of device. Device(s) 610 may represent, but are not limited to, server computers, desktop computers, web-server computers, personal computers, mobile computers, laptop computers, tablet computers, or any other sort of computing device.

A client computing device (e.g., one of client computing device(s) 606(1) through 606(N)) (each of which are also referred to herein as a “data processing system”) may belong to a variety of classes of devices, which may be the same as, or different from, device(s) 610, such as traditional client-type devices, desktop computer-type devices, mobile-type devices, special purpose-type devices, embedded-type devices, and/or wearable-type devices. Thus, a client computing device can include, but is not limited to, a desktop computer, a game console and/or a gaming device, a tablet computer, a personal data assistant (“PDA”), a mobile phone/tablet hybrid, a laptop computer, a telecommunication device, a computer navigation type client computing device such as a satellite-based navigation system including a global positioning system (“GPS”) device, a wearable device, a virtual reality (“VR”) device, an augmented reality (“AR”) device, an implanted computing device, an automotive computer, a network-enabled television, a thin client, a terminal, an Internet of Things (“IoT”) device, a work station, a media player, a personal video recorder (“PVR”),

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a set-top box, a camera, an integrated component (e.g., a peripheral device) for inclusion in a computing device, an appliance, or any other sort of computing device. Moreover, the client computing device may include a combination of the earlier listed examples of the client computing device such as, for example, desktop computer-type devices or a mobile-type device in combination with a wearable device, etc.

Client computing device(s) 606(1) through 606(N) of the various classes and device types can represent any type of computing device having one or more data processing unit(s) 692 operably connected to computer-readable media 694 such as via a bus 616, which in some instances can include one or more of a system bus, a data bus, an address bus, a PCI bus, a Mini-PCI bus, and any variety of local, peripheral, and/or independent buses.

Executable instructions stored on computer-readable media 694 may include, for example, an operating system 619, a client module 620, a profile module 622, and other modules, programs, or applications that are loadable and executable by data processing units(s) 692.

Client computing device(s) 606(1) through 606(N) may also include one or more interface(s) 624 to enable communications between client computing device(s) 606(1) through 606(N) and other networked devices, such as device(s) 610, over network(s) 608. Such network interface(s) 624 may include one or more network interface controllers (NICs) or other types of transceiver devices to send and receive communications and/or data over a network. Moreover, client computing device(s) 606(1) through 606(N) can include input/output (“I/O”) interfaces (devices) 626 that enable communications with input/output devices such as user input devices including peripheral input devices (e.g., a game controller, a keyboard, a mouse, a pen, a voice input device such as a microphone, a video camera for obtaining and providing video feeds and/or still images, a touch input device, a gestural input device, and the like) and/or output devices including peripheral output devices (e.g., a display, a printer, audio speakers, a haptic output device, and the like). FIG. 8 illustrates that client computing device 606(1) is in some way connected to a display device (e.g., a display screen 629(N)), which can display a UI according to the techniques described herein.

In the example environment 600 of FIG. 8, client computing devices 606(1) through 606(N) may use their respective client modules 620 to connect with one another and/or other external device(s) in order to participate in the communication session 603, or in order to contribute activity to a collaboration environment. For instance, a first user may utilize a client computing device 606(1) to communicate with a second user of another client computing device 606(2). When executing client modules 620, the users may share data, which may cause the client computing device 606(1) to connect to the system 602 and/or the other client computing devices 606(2) through 606(N) over the network(s) 608.

The client computing device(s) 606(1) through 606(N) may use their respective profile modules 622 to generate participant profiles (not shown in FIG. 8) and provide the participant profiles to other client computing devices and/or to the device(s) 610 of the system 602. A participant profile may include one or more of an identity of a user or a group of users (e.g., a name, a unique identifier (“ID”), etc.), user data such as personal data, machine data such as location (e.g., an IP address, a room in a building, etc.) and technical capabilities, etc. Participant profiles may be utilized to register participants for communication sessions.

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As shown in FIG. 8, the device(s) 610 of the system 602 include a server module 630 and an output module 632. In this example, the server module 630 is configured to receive, from individual client computing devices such as client computing devices 606(1) through 606(N), media streams 634(1) through 634(N). As described above, media streams can comprise a video feed (e.g., audio and visual data associated with a user), audio data which is to be output with a presentation of an avatar of a user (e.g., an audio only experience in which video data of the user is not transmitted), text data (e.g., text messages), file data and/or screen sharing data (e.g., a document, a slide deck, an image, a video displayed on a display screen, etc.), and so forth. Thus, the server module 630 is configured to receive a collection of various media streams 634(1) through 634(N) during a live viewing of the communication session 603 (the collection being referred to herein as “media data 634”). In some scenarios, not all of the client computing devices that participate in the communication session 603 provide a media stream. For example, a client computing device may only be a consuming, or a “listening”, device such that it only receives content associated with the communication session 603 but does not provide any content to the communication session 603.

In various examples, the server module 630 can select aspects of the media streams 634 that are to be shared with individual ones of the participating client computing devices 606(1) through 606(N). Consequently, the server module 630 may be configured to generate session data 636 based on the streams 634 and/or pass the session data 636 to the output module 632. Then, the output module 632 may communicate communication data 639 to the client computing devices (e.g., client computing devices 606(1) through 606(3) participating in a live viewing of the communication session). The communication data 639 may include video, audio, and/or other content data, provided by the output module 632 based on content 650 associated with the output module 632 and based on received session data 636. The content 650 can include the streams 634 or other shared data, such as an image file, a spreadsheet file, a slide deck, a document, etc. The streams 634 can include a video component depicting images captured by an I/O device 626 on each client computer.

As shown, the output module 632 transmits communication data 639(1) to client computing device 606(1), and transmits communication data 639(2) to client computing device 606(2), and transmits communication data 639(3) to client computing device 606(3), etc. The communication data 639 transmitted to the client computing devices can be the same or can be different (e.g., positioning of streams of content within a user interface may vary from one device to the next).

In various implementations, the device(s) 610 and/or the client module 620 can include GUI presentation module 640. The GUI presentation module 640 may be configured to analyze communication data 639 that is for delivery to one or more of the client computing devices 606. Specifically, the UI presentation module 640, at the device(s) 610 and/or the client computing device 606, may analyze communication data 639 to determine an appropriate manner for displaying video, image, and/or content on the display screen 629 of an associated client computing device 606. In some implementations, the GUI presentation module 640 may provide video, image, and/or content to a presentation GUI 646 rendered on the display screen 629 of the associated client computing device 606. The presentation GUI 646 may be caused to be rendered on the display screen 629 by

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the GUI presentation module 640. The presentation GUI 646 may include the video, image, and/or content analyzed by the GUI presentation module 640.

In some implementations, the presentation GUI 646 may include a plurality of sections or grids that may render or comprise video, image, and/or content for display on the display screen 629. For example, a first section of the presentation GUI 646 may include a video feed of a presenter or individual, a second section of the presentation GUI 646 may include a video feed of an individual consuming meeting information provided by the presenter or individual. The GUI presentation module 640 may populate the first and second sections of the presentation GUI 646 in a manner that properly imitates an environment experience that the presenter and the individual may be sharing.

In some implementations, the GUI presentation module 640 may enlarge or provide a zoomed view of the individual represented by the video feed in order to highlight a reaction, such as a facial feature, the individual had to the presenter. In some implementations, the presentation GUI 646 may include a video feed of a plurality of participants associated with a meeting, such as a general communication session. In other implementations, the presentation GUI 646 may be associated with a channel, such as a chat channel, enterprise Teams channel, or the like. Therefore, the presentation GUI 646 may be associated with an external communication session that is different from the general communication session.

FIG. 9 illustrates a diagram that shows example components of an example device 700 (also referred to herein as a “computing device”) configured to generate data for some of the user interfaces disclosed herein. The device 700 may generate data that may include one or more sections that may render or comprise video, images, virtual objects, and/or content for display on the display screen 629. The device 700 may represent one of the device(s) described herein. Additionally, or alternatively, the device 700 may represent one of the client computing devices 606.

As illustrated, the device 700 includes one or more data processing unit(s) 702, computer-readable media 704, and communication interface(s) 706. The components of the device 700 are operatively connected, for example, via a bus 709, which may include one or more of a system bus, a data bus, an address bus, a PCI bus, a Mini-PCI bus, and any variety of local, peripheral, and/or independent buses.

As utilized herein, data processing unit(s), such as the data processing unit(s) 702 and/or data processing unit(s) 692, may represent, for example, a CPU-type data processing unit, a GPU-type data processing unit, a field-programmable gate array (“FPGA”), another class of DSP, or other hardware logic components that may, in some instances, be driven by a CPU. For example, and without limitation, illustrative types of hardware logic components that may be utilized include Application-Specific Integrated Circuits (“ASICs”), Application-Specific Standard Products (“ASSPs”), System-on-a-Chip Systems (“SOCs”), Complex Programmable Logic Devices (“CPLDs”), etc.

As utilized herein, computer-readable media, such as computer-readable media 704 and computer-readable media 694, may store instructions executable by the data processing unit(s). The computer-readable media may also store instructions executable by external data processing units such as by an external CPU, an external GPU, and/or executable by an external accelerator, such as an FPGA type accelerator, a DSP type accelerator, or any other internal or external accelerator. In various examples, at least one CPU, GPU, and/or accelerator is incorporated in a computing

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device, while in some examples one or more of a CPU, GPU, and/or accelerator is external to a computing device.

Computer-readable media, which might also be referred to herein as a computer-readable medium, may include computer storage media and/or communication media. Computer storage media may include one or more of volatile memory, nonvolatile memory, and/or other persistent and/or auxiliary computer storage media, removable and non-removable computer storage media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Thus, computer storage media includes tangible and/or physical forms of media included in a device and/or hardware component that is part of a device or external to a device, including but not limited to random access memory (“RAM”), static random-access memory (“SRAM”), dynamic random-access memory (“DRAM”), phase change memory (“PCM”), read-only memory (“ROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), flash memory, compact disc read-only memory (“CD-ROM”), digital versatile disks (“DVDs”), optical cards or other optical storage media, magnetic cassettes, magnetic tape, magnetic disk storage, magnetic cards or other magnetic storage devices or media, solid-state memory devices, storage arrays, network attached storage, storage area networks, hosted computer storage or any other storage memory, storage device, and/or storage medium that can be used to store and maintain information for access by a computing device. The computer storage media can also be referred to herein as computer-readable storage media, non-transitory computer-readable storage media, non-transitory computer-readable medium, or computer storage medium.

In contrast to computer storage media, communication media may embody computer-readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave, or other transmission mechanism. As defined herein, computer storage media does not include communication media. That is, computer storage media does not include communications media consisting solely of a modulated data signal, a carrier wave, or a propagated signal, per se.

Communication interface(s) **706** may represent, for example, network interface controllers (“NICs”) or other types of transceiver devices to send and receive communications over a network. Furthermore, the communication interface(s) **706** may include one or more video cameras and/or audio devices **722** to enable generation of video feeds and/or still images, and so forth.

In the illustrated example, computer-readable media **704** includes a data store **708**. In some examples, the data store **708** includes data storage such as a database, data warehouse, or other type of structured or unstructured data storage. In some examples, the data store **708** includes a corpus and/or a relational database with one or more tables, indices, stored procedures, and so forth to enable data access including one or more of hypertext markup language (“HTML”) tables, resource description framework (“RDF”) tables, web ontology language (“OWL”) tables, and/or extensible markup language (“XML”) tables, for example.

The data store **708** may store data for the operations of processes, applications, components, and/or modules stored in computer-readable media **704** and/or executed by data processing unit(s) **702** and/or accelerator(s). For instance, in some examples, the data store **708** may store a data structure **190**, the supplemental data **191**, and session data **636**. The

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session data **636** can include a total number of participants (e.g., users and/or client computing devices) in a communication session, activity that occurs in the communication session, a list of invitees to the communication session, and/or other data related to when and how the communication session is conducted or hosted. The session data may also include contextual data, such as the content that includes video, audio, or other content for rendering and display on one or more of the display screens. The session data may also define permissions that allow or restrict each computer in the system to display select renderings or make any of the user interface transitions disclosed herein.

Hardware data **711** can define aspects of any device, such as a number of display screens of a computer. The contextual data can define any type of activity or status related to the individual users **10A-10L** each associated with individual video streams of a plurality of video streams **634**. For instance, the contextual data can define a person’s level in an organization, how each person’s level relates to the level of others, a performance level of a person, or any other activity or status information that can be used to determine a position for a rendering of a person within a virtual environment. This contextual information can also be fed into any of the models to help bring emphasis to keywords spoken by a person at a specific level, highlight a UI when a background sound of a person at a certain level is detected, or change a sentiment display in a particular way when a person at a certain level is detected has a certain sentiment.

Alternately, some or all of the above-referenced data can be stored on separate memories **716** on board one or more data processing unit(s) **702** such as a memory on board a CPU-type processor, a GPU-type processor, an FPGA-type accelerator, a DSP-type accelerator, and/or another accelerator. In this example, the computer-readable media **704** also includes an operating system **718** and application programming interface(s) **710** (APIs) configured to expose the functionality and the data of the device **700** to other devices. Additionally, the computer-readable media **704** includes one or more modules such as the server module **730**, the output module **732**, and the GUI presentation module **740**, although the number of illustrated modules is just an example, and the number may vary. That is, functionality described herein in association with the illustrated modules may be performed by a fewer number of modules or a larger number of modules on one device or spread across multiple devices.

In closing, although the various configurations have been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended representations is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed subject matter.

I claim:

1. A method, executed by a computing system, for promoting engagement by transitioning to a temporary viewing perspective from a first viewing perspective associated with a first entity that is one of a plurality of entities positioned within a first 3D environment, comprising:

accessing a first model defining the first 3D environment, the first model defines a position and an orientation for each entity of the plurality of entities of the 3D environment, the first model defines the first viewing perspective that originates from a point of view of the first entity;

generating a first user interface arrangement comprising a rendering of the first 3D environment using first model

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comprising the plurality of entities, wherein the rendering of the first 3D environment is from the first viewing perspective associated with the first entity for display on a display device associated with the first user;

determining that an input identifying an event meets a preset condition, wherein the input is for causing a transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising a subset of entities that is selected from the plurality of entities;

in response to determining that the input identifying the event meets the preset condition, wherein the input is for causing the transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising the subset of entities that is selected from the plurality of entities:

selecting the subset of entities from the plurality of entities, wherein the subset of entities is selected for being associated with the event that meets the preset condition;

generating the second model defining a second 3D environment comprising the subset of entities selected from the plurality of entities from the first 3D environment, wherein the subset of entities are each selected in response to determining that each of the subset of entities are associated with the event, wherein other entities of the plurality of entities that are not associated with the event are excluded from the second model defining the second 3D environment, wherein the subset of entities are positioned within the temporary viewing perspective of a virtual camera, and

causing a transition from the first user interface arrangement of the rendering of the first 3D environment using the first model to a second user interface arrangement of the second 3D environment using the second model, the second user interface arrangement including a rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera for display on the display device associated with the first user, wherein the rendering of the subset of entities is displayed using the second model defining the second 3D environment comprising the subset of entities each representing the users associated with the event.

2. The method of claim 1, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users have a threshold level of affiliation with one another, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals from the subset of computing devices associated with the subset of users having the threshold level of affiliation with one another.

3. The method of claim 1, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users and the activity signals indicate a threshold level of activity, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity

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signals from the subset of computing devices associated with the subset of users, wherein the activity signals indicate the threshold level of activity.

4. The method of claim 3, wherein the rendering of the subset of entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset of users reduces below a second activity threshold level, in response to the activity signals of the subset of users reducing below the second activity threshold level, the display on the display device associated with the first user returns to the rendering of the 3D environment using the first viewing perspective associated with the first entity.

5. The method of claim 1, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users have a threshold level of affiliation with one another, and where the activity signals indicate the threshold level of activity, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals indicating the threshold level of activity from the subset of computing devices associated with the subset of users having the threshold level of affiliation with one another.

6. The method of claim 5, wherein the rendering of the subset of entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset of users reduces below a second activity threshold level, in response to the activity signals of the subset of users reducing below the second activity threshold level, the display on the display device associated with the first user returns to the rendering of the 3D environment using the first viewing perspective associated with the first entity.

7. The method of claim 1, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users are identified on an organizational chart as having an affiliation, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals from the subset of computing devices associated with the subset of users are identified on the organizational chart as having the affiliation.

8. The method of claim 1, wherein permissions are modified in response to the input identifying the event meets the preset condition, wherein the modified permission allow a first set of computing devices of the users associated with the event to display the rendering of the subset of entities using the supplemental model defining the second 3D environment, wherein the permissions are modified in response to the event to restrict a second set of computers associated with other users from displaying the rendering of the subset of entities using the supplemental model defining the second 3D environment.

9. The method of claim 1, wherein the computing system is in a first operating state for displaying the first user interface arrangement while accessing the first model defining the first 3D environment, while in the first operating state the computing system is configured to receive signals from devices of the plurality of entities indicating the event, wherein the computing system is in a second operating state

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for displaying the second user interface arrangement while accessing the second model defining the second 3D environment.

10. A computing device for promoting engagement by transitioning to a temporary viewing perspective from a first viewing perspective associated with a first entity that is one of a plurality of entities positioned within a first 3D environment, the computing device comprising:

one or more processing units; and

a computer-readable storage medium having encoded thereon computer-executable instructions to cause the one or more processing units to:

access a first model defining the first 3D environment, the first model defines a position and an orientation for each entity of the plurality of entities of the 3D environment, the first model defines the first viewing perspective that originates from a point of view of the first entity;

generate a first user interface arrangement comprising a rendering of the first 3D environment using first model comprising the plurality of entities, wherein the rendering of the first 3D environment is from the first viewing perspective associated with the first entity for display on a display device associated with the first user;

determine that an input identifying an event meets a preset condition, wherein the input is for causing a transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising a subset of entities that is selected from the plurality of entities;

in response to determining that the input identifying the event meets the preset condition, wherein the input is for causing the transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising the subset of entities that is selected from the plurality of entities:

select the subset of entities from the plurality of entities, wherein the subset of entities is selected for being associated with the event that meets the preset condition;

generate the second model defining a second 3D environment comprising the subset of entities selected from the plurality of entities from the first 3D environment, wherein the subset of entities are each selected in response to determining that each of the subset of entities are associated with the event, wherein other entities of the plurality of entities that are not associated with the event are excluded from the second model defining the second 3D environment, wherein the subset of entities are positioned within the temporary viewing perspective of a virtual camera, and

cause a transition from the first user interface arrangement of the rendering of the first 3D environment using the first model to a second user interface arrangement of the second 3D environment using the second model, the second user interface arrangement including a rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera for display on the display device associated with the first user, wherein the rendering of the subset of entities is displayed using the second model defining the second 3D environment compris-

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ing the subset of entities each representing the users associated with the event.

11. The computing device of claim 10, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users have a threshold level of affiliation with one another, wherein the generation of the second model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals from the subset of computing devices associated with the subset of users having the threshold level of affiliation with one another.

12. The computing device of claim 10, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users and the activity signals indicate a threshold level of activity, wherein the generation of the second model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals from the subset of computing devices associated with the subset of users, wherein the activity signals indicate the threshold level of activity.

13. The computing device of claim 12, wherein the rendering of the subset of entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset of users reduces below a second activity threshold level, in response to the activity signals of the subset of users reducing below the second activity threshold level, the display on the display device associated with the first user returns to the rendering of the 3D environment using the first viewing perspective associated with the first entity.

14. The computing device of claim 10, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users have a threshold level of affiliation with one another, and where the activity signals indicate the threshold level of activity, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals indicating the threshold level of activity from the subset of computing devices associated with the subset of users having the threshold level of affiliation with one another.

15. The computing device of claim 14, wherein the rendering of the subset of entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset of users reduces below a second activity threshold level, in response to the activity signals of the subset of users reducing below the second activity threshold level, the display on the display device associated with the first user returns to the rendering of the 3D environment using the first viewing perspective associated with the first entity.

16. The computing device of claim 10, wherein permissions are modified in response to the input identifying the event meets the preset condition, wherein the modified permission allow computing devices of the users associated with the event to display the rendering of the subset of entities using the supplemental model defining the second 3D environment.

17. A computer-readable storage device having encoded thereon computer-executable instructions for promoting engagement by transitioning to a temporary viewing per-

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spective from a first viewing perspective associated with a first entity that is one of a plurality of entities positioned within a 3D environment, the computer-executable instructions to cause the one or more processing units of a computing device to:

access a first model defining the first 3D environment, the first model defines a position and an orientation for each entity of the plurality of entities of the 3D environment, the first model defines the first viewing perspective that originates from a point of view of the first entity;

generate a first user interface arrangement comprising a rendering of the first 3D environment using first model comprising the plurality of entities, wherein the rendering of the first 3D environment is from the first viewing perspective associated with the first entity for display on a display device associated with the first user;

determine that an input identifying an event meets a preset condition, wherein the input is for causing a transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising a subset of entities that is selected from the plurality of entities;

in response to determining that the input identifying the event meets the preset condition, wherein the input is for causing the transition from the first user interface arrangement using the first model comprising a plurality of entities to a second user interface arrangement using a second model comprising the subset of entities that is selected from the plurality of entities:

select the subset of entities from the plurality of entities, wherein the subset of entities is selected for being associated with the event that meets the preset condition;

generate the second model defining a second 3D environment comprising the subset of entities selected from the plurality of entities from the first 3D environment, wherein the subset of entities are each selected in response to determining that each of the subset of entities are associated with the event, wherein other entities of the plurality of entities that are not associated with the event are excluded from the second model defining the second 3D environ-

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ment, wherein the subset of entities are positioned within the temporary viewing perspective of a virtual camera, and

cause a transition from the first user interface arrangement of the rendering of the first 3D environment using the first model to a second user interface arrangement of the second 3D environment using the second model, the second user interface arrangement including a rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera for display on the display device associated with the first user, wherein the rendering of the subset of entities is displayed using the second model defining the second 3D environment comprising the subset of entities each representing the users associated with the event.

18. The computer-readable storage device of claim 17, wherein the input identifying the event meets the preset condition when activity signals are received from a subset of computing devices associated with a subset of users and the activity signals indicate a threshold level of activity, wherein the generation of the supplemental model and the rendering of the subset of entities positioned within the temporary viewing perspective of the virtual camera is in response to receiving the activity signals from the subset of computing devices associated with the subset of users, wherein the activity signals indicate the threshold level of activity.

19. The computer-readable storage device of claim 17, wherein the rendering of the subset of entities positioned within the viewing perspective of a virtual camera are displayed on the display device until the activity signals of the subset of users reduces below a second activity threshold level, in response to the activity signals of the subset of users reducing below the second activity threshold level, the display on the display device associated with the first user returns to the rendering of the 3D environment using the first viewing perspective associated with the first entity.

20. The computer-readable storage device of claim 17, wherein permissions are modified in response to the input identifying the event meets the preset condition, wherein the modified permission allow computing devices of the users associated with the event to display the rendering of the subset of entities using the supplemental model defining the second 3D environment.

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